“Phase I” method for initial feasible point (0.25 pt)

We consider the original constraints:

$$
\begin{aligned}
&(\mathrm{C1})\ x - 2y + 2 \geq 0,\\
&(\mathrm{C2})\ -x - 2y + 6 \geq 0,\\
&(\mathrm{C3})\ -x + 2y + 2 \geq 0,\\
&(\mathrm{C4})\ x \geq 0,\\
&(\mathrm{C5})\ y \geq 0.
\end{aligned}
$$

Given the starting point

$$x_0 = \begin{pmatrix} 1 \\ 2.5 \end{pmatrix},$$

we first check feasibility:

$$
\begin{aligned}
&\mathrm{C1:}\ 1 - 2\cdot 2.5 + 2 = -2 \quad \text{(violated by 2)},\\
&\mathrm{C2:}\ -1 - 2\cdot 2.5 + 6 = 0 \quad \text{(active)},\\
&\mathrm{C3:}\ -1 + 2\cdot 2.5 + 2 = 6 \quad (> 0),\\
&\mathrm{C4:}\ 1 \geq 0, \qquad \mathrm{C5:}\ 2.5 \geq 0.
\end{aligned}
$$

Only Constraint (C1) is violated.

Phase I reformulation

Introduce the slack variable $s \geq 0$ to account for infeasibility and modify the constraints to follow the convention used in the lecture:

$$
\begin{aligned}
\min_{x,y,s} \quad & s\\
\text{s.t.} \quad & -x + 2y - 2 \leq s,\\
& x + 2y - 6 \leq 0,\\
& x - 2y - 2 \leq 0,\\
& -x \leq 0, \quad -y \leq 0, \quad s \geq 0.
\end{aligned}
$$

Choose $s_0 = 2 = \max(0, g_1(x_0))$ so that Constraint (C1) becomes satisfied at the starting point:

$$-1 + 2\cdot 2.5 - 2 = 2.$$

Thus $(x_0, y_0, s_0) = (1,\ 2.5,\ 2)$ is feasible for the Phase I problem.

Solving Phase I

Because the original problem is feasible, the minimum of Phase I is $s^\star = 0$, attained at any point satisfying (C1)–(C5).

A suitable feasible point is found by intersecting (C1) and (C2):

$$
\begin{cases}
-x + 2y - 2 = 0,\\
x + 2y - 6 = 0.
\end{cases}
$$

Solve the system:

$$x = 2y - 2,$$

$$(2y - 2) + 2y - 6 = 0 \ \Rightarrow\ 4y - 8 = 0 \ \Rightarrow\ y = 2,$$

$$x = 2 \cdot 2 - 2 = 2.$$

Thus the candidate point is:

$$(x, y, s) = (2, 2, 0).$$

Check feasibility:

$$\begin{aligned}
&\text{C1: } 2 - 2 \cdot 2 + 2 = 0 \geq 0, \\
&\text{C2: } -2 - 2 \cdot 2 + 6 = 0 \geq 0, \\
&\text{C3: } -2 + 2 \cdot 2 + 2 = 4 \geq 0, \\
&\text{C4: } 2 \geq 0, \qquad \text{C5: } 2 \geq 0, \\
&\quad s = 0.
\end{aligned}$$

Thus, a valid feasible point for the original problem, produced by the Phase I method, is

$$x_{\text{feasible}} = \begin{pmatrix} 2 \\ 2 \end{pmatrix},$$

which satisfies all original constraints and can be used to start the active-set method.

"Big M" method for initial feasible point (Bonus 0.5 pt)

We introduce the variable

$$z := \begin{pmatrix} x \\ y \\ s \end{pmatrix} \in \mathbb{R}^3$$

and rewrite the Big M formulation as a quadratic program of the form

$$\min_z \ \frac{1}{2} z^\top Q z + q^\top z + r \quad \text{subject to} \quad Bz \leq b.$$

The Big M objective is

$$F_M(x, y, s) = (x-1)^2 + (y-2.5)^2 + 100\, s,$$

so

$$Q = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad q = \begin{pmatrix} -2 \\ -5 \\ 100 \end{pmatrix}, \qquad r = 7.25.$$

The constraints

$$\begin{aligned}
&-x + 2y - 2 \leq s, \\
&x + 2y - 6 \leq 0, \\
&x - 2y - 2 \leq 0, \\
&-x \leq 0, \quad -y \leq 0, \quad s \geq 0
\end{aligned}$$

are written as

$$Bz \leq b, \quad B \in \mathbb{R}^{6\times 3},\ b \in \mathbb{R}^6,$$

with rows

$$B = \begin{pmatrix} -1 & 2 & -1 \\ 1 & 2 & 0 \\ 1 & -2 & 0 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \qquad b = \begin{pmatrix} 2 \\ 6 \\ 2 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

The starting point is

$$z_0 = \begin{pmatrix} 1 \\ 2.5 \\ 2 \end{pmatrix},$$

for which

$$Bz_0 - b = \begin{pmatrix} 0 \\ 0 \\ -6 \\ -1 \\ -2.5 \\ -2 \end{pmatrix} \leq 0,$$

so z_0 is feasible and constraints 1 and 2 are active:

$$\mathcal{A}_0 = \{1, 2\}.$$

Active-set method for the Big M QP

For a given active set $\mathcal{A}_k$, let B_k and b_k collect the corresponding rows of B and entries of b. The active-set subproblem is

$$\min_z \frac{1}{2} z^\top Q z + q^\top z + r \quad \text{s.t.} \quad B_k z = b_k,$$

whose KKT system is

$$\begin{pmatrix} Q & B_k^\top \\ B_k & 0 \end{pmatrix} \begin{pmatrix} z_{k+1}^{\text{cand}} \\ \lambda_{k+1,\mathcal{A}_k} \end{pmatrix} = \begin{pmatrix} -q \\ b_k \end{pmatrix}.$$

This gives a candidate point z_{k+1}^{cand}; the search direction is $p_k = z_{k+1}^{\text{cand}} - z_k$, and we choose a step length $\alpha_k \in (0, 1]$ so that $z_k + \alpha_k p_k$ remains feasible. If $\alpha_k < 1$, a new constraint becomes active. If $p_k = 0$, we inspect the multipliers and possibly drop a constraint with negative multiplier.

Iteration 0. Active set: $\mathcal{A}_0 = \{1, 2\}$, so

$$B_0 = \begin{pmatrix} -1 & 2 & -1 \\ 1 & 2 & 0 \end{pmatrix}, \qquad b_0 = \begin{pmatrix} 2 \\ 6 \end{pmatrix}.$$

The KKT system

$$\begin{pmatrix} Q & B_0^\top \\ B_0 & 0 \end{pmatrix} \begin{pmatrix} z_1^{\text{cand}} \\ \lambda_{1,\mathcal{A}_0} \end{pmatrix} = \begin{pmatrix} -q \\ b_0 \end{pmatrix}$$

yields

$$z_1^{\text{cand}} = \begin{pmatrix} 81 \\ -37.5 \\ -158 \end{pmatrix}, \qquad \lambda_{1,1} = 100,\ \lambda_{1,2} = -60.$$

The search direction is

$$p_0 = z_1^{\text{cand}} - z_0 = \begin{pmatrix} 80 \\ -40 \\ -160 \end{pmatrix}.$$

The inactive constraints along $z(\alpha) = z_0 + \alpha p_0$ satisfy

$$\begin{pmatrix} 160\alpha - 6 \\ -80\alpha - 1 \\ 40\alpha - 2.5 \\ 160\alpha - 2 \end{pmatrix} \leq 0,$$

which gives

$$\alpha_0 \le \min\left\{\frac{6}{160}, \frac{2.5}{40}, \frac{2}{160}\right\} = \frac{1}{80}.$$

The blocking constraint is 6, so $\alpha_0 = 1/80$ and

$$z_1 = z_0 + \alpha_0 p_0 = \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix}.$$

At z_1, $Bz_1 - b = (0, 0, -4, -2, -2, 0)^\top$, so

$$\mathcal{A}_1 = \{1, 2, 6\}.$$

Iteration 1. With $\mathcal{A}_1 = \{1, 2, 6\}$,

$$B_1 = \begin{pmatrix} -1 & 2 & -1 \\ 1 & 2 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \qquad b_1 = \begin{pmatrix} 2 \\ 6 \\ 0 \end{pmatrix}.$$

Solving

$$\begin{pmatrix} Q & B_1^\top \\ B_1 & 0 \end{pmatrix} \begin{pmatrix} z_2^{\text{cand}} \\ \lambda_{2,\mathcal{A}_1} \end{pmatrix} = \begin{pmatrix} -q \\ b_1 \end{pmatrix}$$

gives

$$z_2^{\text{cand}} = \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix} = z_1, \qquad \lambda_{2,1} = \tfrac{5}{4},\ \lambda_{2,2} = -\tfrac{3}{4},\ \lambda_{2,6} = \tfrac{395}{4}.$$

Hence $p_1 = 0$. Since one multiplier is negative ($\lambda_{2,2} < 0$), constraint 2 is removed from the active set:

$$\mathcal{A}_2 = \{1, 6\}, \qquad z_2 := z_1.$$

Iteration 2. Now

$$B_2 = \begin{pmatrix} -1 & 2 & -1 \\ 0 & 0 & -1 \end{pmatrix}, \qquad b_2 = \begin{pmatrix} 2 \\ 0 \end{pmatrix}.$$

The KKT system

$$\begin{pmatrix} Q & B_2^\top \\ B_2 & 0 \end{pmatrix} \begin{pmatrix} z_3^{\text{cand}} \\ \lambda_{3,\mathcal{A}_2} \end{pmatrix} = \begin{pmatrix} -q \\ b_2 \end{pmatrix}$$

yields

$$z_3^{\text{cand}} = \begin{pmatrix} \frac{7}{5} \\ \frac{17}{10} \\ 0 \end{pmatrix}, \qquad \lambda_{3,1} = \frac{4}{5},\ \lambda_{3,6} = \frac{496}{5}.$$

The search direction is

$$p_2 = z_3^{\text{cand}} - z_2 = \begin{pmatrix} -\frac{3}{5} \\ -\frac{3}{10} \\ 0 \end{pmatrix}.$$

Along $z(\alpha) = z_2 + \alpha p_2$, the inactive constraints satisfy

$$\begin{pmatrix} -\frac{6}{5}\alpha \\ -4 \\ \frac{3}{5}\alpha - 2 \\ \frac{3}{10}\alpha - 2 \end{pmatrix} \le 0$$

for all $\alpha \in [0,1]$. Thus no new constraint blocks the step and we take $\alpha_2 = 1$:

$$z_3 = z_2 + p_2 = \begin{pmatrix} \frac{7}{5} \\ \frac{17}{10} \\ 0 \end{pmatrix}.$$

At z_3, $Bz_3 - b = (0, -\frac{6}{5}, -4, -\frac{7}{5}, -\frac{17}{10}, 0)^\top$, so

$$\mathcal{A}_3 = \{1, 6\}.$$

Iteration 3 (optimality). Solving the KKT system again with $\mathcal{A}_3 = \{1, 6\}$ returns

$$z_4^{\text{cand}} = z_3, \qquad \lambda_{4,1} = \frac{4}{5},\ \lambda_{4,6} = \frac{496}{5},$$

so $p_3 = 0$ and all multipliers of active constraints are nonnegative. Therefore the KKT conditions are satisfied and the active-set method terminates at

$$(x^\star, y^\star, s^\star) = \left(\frac{7}{5}, \frac{17}{10}, 0\right) = (1.4, 1.7, 0)\,.$$

This matches the solution we found in the first exercise as expected.